GRADE 1 · BLOCK CODING

Order Matters

Same blocks, different order, different number

Name: .... Date: ....

★ **I can show how the order of the blocks changes the result.**

Where the path stops

Bit's path stops at 5. If a forward block would go past 5, Bit just stops on 5. (Bit also cannot go below 0.) That is why the ORDER of the blocks can change where Bit lands.

Bit's number path

0	1	2	3	4	5

Code 1

start at 3

forward 3

back 1

1 Run Code 1

Start on 3. Forward 3 would pass the end, so Bit stops on 5. Then back 1. What number does Bit land on? Write it.

Code 2 — same blocks, new order

start at 3

back 1

forward 3

2 Predict, then run Code 2

PREDICT first: where will Bit land? Then run Code 2 from 3: back 1, then forward 3. What number does Bit land on?

3 You try — think and build

(a) Code 1 and Code 2 use the SAME blocks. Did Bit land on the same number? Circle YES or NO, and tell why. (b) Put these three blocks in an order that gets Bit to 5.

Big idea

The blocks run in order from the top. The same blocks in a different order can give a different result — so the order of your code matters.

TEACHER GUIDE · GRADE 1 · BLOCK CODING

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Quick facilitation notes & answer key

What this worksheet teaches

Bit moves on a number line from 0 to 5. Students run two sets of instructions made of the SAME blocks in a different order — and find Bit lands on different numbers. Then they explain why and build their own order to hit a target. No coding experience needed.

Before you start (no computer needed)

Paper and pencil only. Optional: tape a number line 0 to 5 on the floor so students can walk both codes and feel Bit stop at the end (5).

How to lead it (about 20 minutes)

1. Show them first (you do). Walk Code 1 aloud: "start at 3… forward 3 would reach 6, but the path stops at 5, so Bit stops on 5… back 1 to 4." One block at a time.

2. Do one together (we do). Exercise 1: Code 1 lands on 4. Exercise 2: have students predict Code 2, then run it — back 1 to 2, forward 3 to 5.

3. Let them try alone (you do). Exercise 3: the same blocks landed on different numbers (4 vs 5); students explain why, then write an order that reaches 5.

4. Big idea: "The computer follows instructions in the exact order you give. Change the order and you can change what happens."

Answer key (these are the verified correct answers)

Exercise 1 — Code 1: start 3, forward 3 stops at 5 (end of the path), back 1 → 4. Bit lands on 4.

Exercise 2 — Code 2: start 3, back 1 → 2, forward 3 → 5. Bit lands on 5.

Exercise 3 — (a) NO, not the same (4 vs 5). Why: in Code 1 the forward 3 runs into the end of the path, so Bit stops at 5 and loses a step; in Code 2 Bit steps back first, so the forward 3 fits. (b) Any order that reaches 5, e.g. start at 3, back 1, forward 3.

If students get stuck — and ways to adjust

Watch for: expecting 3 + 3 – 1 to always be 5. The edge of the path is what makes the order matter — show it on the floor line.

Make it easier: walk Code 1 and Code 2 side by side on the floor line so the different landings are felt, not just calculated.

Make it harder: ask for an order that lands Bit on 4, and whether more than one order can reach the same number.

Ontario curriculum link (official wording)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential events.

In plain terms: students follow (or write) step-by-step instructions, in order, to get a result.

C3.2 — read and alter existing code, including code that involves sequential events, and describe how changes to the code affect the outcomes.

In plain terms: students read instructions someone wrote, change one, and explain what the change did.

You'll know it worked when

The student lands Code 1 on 4 and Code 2 on 5 and explains that the order changed the result.